

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING



Report on

Analyze the IMDB Movies dataset using Hadoop

MapReduce and Apache Pig

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1. Problem Statement

Analyze the IMDB Movies dataset (imdbmovies.csv) using Hadoop MapReduce and Apache Pig to extract insights such as top-rated movies, genre popularity, prolific directors, and rating trends across decades.

1.1 Dataset Description

1.2 Dataset Description

Column (in file)	Type	Description
Title_ID	string	Unique IMDb-style ID (e.g. tt0000001). 2,000,000 unique values, no nulls.
Title	string	Title of the movie. 1,596,761 unique values; 7 missing.
Title_Type	string	Category of title: short, movie, video, tvMovie, tvSpecial, videoGame.
Year	int	Release year. Range 0–2032; 0 appears to represent ‘unknown’. No nulls.
Genres	string	One or more genres, comma-separated, quoted when multiple. 2,111 unique combinations.
Runtime	int (not float)	Runtime in minutes. Stored as 0, not blank, when unknown (736,857 rows = 36.8%).
Rating	float	IMDb-style rating, 0–10. Stored as 0.0, not blank, when unrated. Median value is 0.
Votes	int	Number of votes backing the rating.
Director_ID	string	IMDb-style director ID (e.g. nm0005690).
Director_Name	string	Director name(s), comma-separated if more than one. 2 missing values.

1.3 Summary of Data Analysis

Following analysis can be performed on the data-set:

1. Total Movies by Genre

2. Movie Count per Director
3. Top 10 Highest Rated Movies
4. Decade-wise Average Rating
5. Year-wise Title Count

Following Pig analysis is performed on the data-set:

1. Movie Count Per Genre
2. Top Directors by Movie Count
3. Top 10 Highest-Rated Movies
4. Average Rating per Genre
5. Average Rating & Release Count per Year
6. Content Count by Title Type
7. Bottom 20 Lowest-Rated Titles

2. Map Reduce Analysis

2.1 Job 1: Total Movies by Genre

The total number of titles is calculated for each genre to determine the distribution of movies across different genres.

Driver Class: (GenreDriver.java)

```
package genrecount;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class GenreDriver {
    public static void main(String[] args) throws Exception {
        if (args.length != 2) {
            System.err.println("Usage: GenreDriver <input> <output>");
            System.exit(-1);
        }
        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "Genre Wise Title Count");
```

```

job.setJarByClass(GenreDriver.class);
// Mapper
job.setMapperClass(GenreMapper.class);
// Reducer
job.setReducerClass(GenreReducer.class);
// Mapper output
job.setMapOutputKeyClass(Text.class);
job.setMapOutputValueClass(IntWritable.class);
// Final output
job.setOutputKeyClass(Text.class);
job.setOutputValueClass(IntWritable.class);
// Input
FileInputFormat.addInputPath(job, new Path(args[0]));
// Output
FileOutputFormat.setOutputPath(job, new Path(args[1]));
System.exit(job.waitForCompletion(true) ? 0 : 1);
}
}

```

Mapper Class: (GenreMapper.java)

```

package genrecount;
import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class GenreMapper
    extends Mapper<LongWritable, Text, Text, IntWritable> {
    private final Text genreKey = new Text();
    private final IntWritable one = new IntWritable(1);
    @Override
    public void map(LongWritable key, Text value, Context context)
        throws IOException, InterruptedException {
        String line = value.toString();
        // Skip header
        if (line.startsWith("Title_ID")) {
            return;
        }
        String[] fields = line.split("\t", -1);
        // Need at least 5 columns
        if (fields.length < 5) {
            return;
        }
        // Column 4 = Genres
        String genres = fields[4];
        if (genres.equals("Unknown") || genres.equals("\N")) {
            return;
        }
        // Example: Action,Adventure,Drama
        String[] genreList = genres.split(",");
        for (String genre : genreList) {
            genre = genre.trim();
            if (!genre.isEmpty()) {
                genreKey.set(genre);
                context.write(genreKey, one);
            }
        }
    }
}

```

```
}  
}
```

Reducer Class: (GenreReducer.java)

```
package genrecount;  
import java.io.IOException;  
import org.apache.hadoop.io.IntWritable;  
import org.apache.hadoop.io.Text;  
import org.apache.hadoop.mapreduce.Reducer;  
  
public class GenreReducer  
    extends Reducer<Text, IntWritable, Text, IntWritable> {  
    private final IntWritable result = new IntWritable();  
    @Override  
    public void reduce(Text key, Iterable<IntWritable> values,  
        Context context)  
        throws IOException, InterruptedException {  
        int total = 0;  
        for (IntWritable value : values) {  
            total += value.get();  
        }  
        result.set(total);  
        context.write(key, result);  
    }  
}
```

EXECUTION

```
C:\Ashfak\IMDBMoviesAnalysis\genrecount>jar -cvf genrecount.jar -C classes .
added manifest
adding: genrecount/(in = 0) (out= 0)(stored 0%)
adding: genrecount/GenreDriver.class(in = 1656) (out= 901)(deflated 45%)
adding: genrecount/GenreMapper.class(in = 2214) (out= 994)(deflated 55%)
adding: genrecount/GenreReducer.class(in = 1783) (out= 719)(deflated 57%)

C:\Ashfak\IMDBMoviesAnalysis\genrecount>dir genrecount.jar
Volume in drive C has no label.
Volume Serial Number is 66AE-5995

Directory of C:\Ashfak\IMDBMoviesAnalysis\genrecount

09/13/2026  05:11 PM                3,500 genrecount.jar
               1 File(s)                3,500 bytes
               0 Dir(s) 190,767,312,896 bytes free

C:\Ashfak\IMDBMoviesAnalysis\genrecount>hdfs dfs -rm -r -f /imdb_movies/output/genrecount

C:\Ashfak\IMDBMoviesAnalysis\genrecount>hadoop jar genrecount.jar genrecount.GenreDriver /imdb_movies/input/imdb_movies.txt /imdb_movies/output/genrecount
2026-09-13 17:13:12,487 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-13 17:13:14,243 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-13 17:13:14,302 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/sudipto/.staging/job_1789296062547_0001
2026-09-13 17:13:14,962 INFO input.FileInputFormat: Total input files to process : 1
2026-09-13 17:13:15,157 INFO mapreduce.JobSubmitter: number of splits:2
2026-09-13 17:13:15,519 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789296062547_0001
2026-09-13 17:13:15,525 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-13 17:13:15,970 INFO conf.Configuration: resource-types.xml not found
2026-09-13 17:13:15,970 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-13 17:13:16,802 INFO impl.YarnClientImpl: Submitted application application_1789296062547_0001
2026-09-13 17:13:16,895 INFO mapreduce.Job: The url to track the job: http://DESKTOP-AD7D20Q:8088/proxy/application_1789296062547_0001/
2026-09-13 17:13:16,895 INFO mapreduce.Job: Running job: job_1789296062547_0001
2026-09-13 17:13:37,410 INFO mapreduce.Job: Job job_1789296062547_0001 running in uber mode : false
2026-09-13 17:13:37,410 INFO mapreduce.Job:  map 0% reduce 0%
2026-09-13 17:14:01,367 INFO mapreduce.Job:  map 100% reduce 0%
2026-09-13 17:14:13,556 INFO mapreduce.Job:  map 100% reduce 100%
2026-09-13 17:14:13,600 INFO mapreduce.Job: Job job_1789296062547_0001 completed successfully
2026-09-13 17:14:13,840 INFO mapreduce.Job: Counters: 55
```

execution1

```
Killed map tasks=1
Launched map tasks=3
Launched reduce tasks=1
Data-local map tasks=3
Total time spent by all maps in occupied slots (ms)=40356
Total time spent by all reduces in occupied slots (ms)=14483
Total time spent by all map tasks (ms)=40356
Total time spent by all reduce tasks (ms)=14483
Total vcore-milliseconds taken by all map tasks=40356
Total vcore-milliseconds taken by all reduce tasks=14483
Total megabyte-milliseconds taken by all map tasks=41324544
Total megabyte-milliseconds taken by all reduce tasks=14830592
Map-Reduce framework
Map input records=2000001
Map output records=3317381
Map output bytes=37363938
Map output materialized bytes=43998712
Input split bytes=240
Combine input records=0
Combine output records=0
Reduce input groups=28
Reduce shuffle bytes=43998712
Reduce input records=3317381
Reduce output records=28
Spilled Records=6634762
Shuffled Maps =2
Failed Shuffles=0
Merged Map outputs=2
GC time elapsed (ms)=566
CPU time spent (ms)=19900
Physical memory (bytes) snapshot=1111920640
Virtual memory (bytes) snapshot=1416998912
Total committed heap usage (bytes)=909639680
Peak Map Physical memory (bytes)=467787776
Peak Map Virtual memory (bytes)=569204736
Peak Reduce Physical memory (bytes)=2289099568
Peak Reduce Virtual memory (bytes)=289599488
Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
File Input Format Counters
Bytes Read=172948614
File Output Format Counters
Bytes Written=389
```

Execution2

OUTPUT

```
C:\Ashfak\IMDBMoviesAnalysis\genrecount>hdfs dfs -cat /imdb_movies/output/genrecount/part-r-00000
Action 104453
Adult 111093
Adventure 60896
Animation 79231
Biography 34998
Comedy 332549
Crime 65922
Documentary 341620
Drama 540219
Family 55728
Fantasy 53753
Film-Noir 877
Game-Show 1662
History 28841
Horror 105914
Music 76069
Musical 22786
Mystery 41586
News 9547
Reality-TV 8962
Romance 90446
Sci-Fi 46625
Short 951588
Sport 32764
Talk-Show 5642
Thriller 83960
War 14838
Western 14812

C:\Ashfak\IMDBMoviesAnalysis\genrecount>
```

output

2.2 Job 2: Movie Count per Director

Driver Class: (DirectorDriver.java)

```
package directorcount;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class DirectorDriver {
    public static void main(String[] args) throws Exception {
        if (args.length != 2) {
            System.err.println("Usage: DirectorDriver <input> <output>");
            System.exit(-1);
        }
        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "Director Wise Title Count");
        job.setJarByClass(DirectorDriver.class);
        // Mapper
        job.setMapperClass(DirectorMapper.class);
        // Reducer
        job.setReducerClass(DirectorReducer.class);
        // Mapper output
        job.setMapOutputKeyClass(Text.class);
        job.setMapOutputValueClass(IntWritable.class);
    }
}
```

```
// Final output
job.setOutputKeyClass(Text.class);
job.setOutputValueClass(IntWritable.class);
// Input
FileInputFormat.addInputPath(job, new Path(args[0]));
// Output
FileOutputFormat.setOutputPath(job, new Path(args[1]));
System.exit(job.waitForCompletion(true) ? 0 : 1);
}
}
```

Mapper Class: (DirectorMapper.java)

```
package directorcount;
import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class DirectorMapper
    extends Mapper<LongWritable, Text, Text, IntWritable> {
    private final Text directorKey = new Text();
    private final IntWritable one = new IntWritable(1);
    @Override
    public void map(LongWritable key, Text value, Context context)
        throws IOException, InterruptedException {
        String line = value.toString();
        // Skip header
        if (line.startsWith("Title_ID")) {
            return;
        }
        String[] fields = line.split("\t", -1);
        // Need at least 10 columns
        if (fields.length < 10) {
            return;
        }
        // Column 9 = Director_Name
        String director = fields[9].trim();
        // Ignore unknown directors
        if (director.isEmpty()
            || director.equals("Unknown")
            || director.equals("\\N")) {
            return;
        }
        directorKey.set(director);
        context.write(directorKey, one);
    }
}
```

Reducer Class: (DirectorReducer.java)

```
package directorcount;
import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class DirectorReducer
```

```
extends Reducer<Text, IntWritable, Text, IntWritable> {  
    private final IntWritable result = new IntWritable();  
    @Override  
    public void reduce(Text key, Iterable<IntWritable> values,  
        Context context)  
        throws IOException, InterruptedException {  
        int total = 0;  
        for (IntWritable value : values) {  
            total += value.get();  
        }  
        result.set(total);  
        context.write(key, result);  
    }  
}
```

EXECUTION

```

Select Administrator: Command Prompt
C:\Ashfak\IMDBMoviesAnalysis\directorcount\hadoop jar directorcount.jar directorcount.DirectorDriver /imdb_movies/input/imdb_movies.txt /imdb_movies/output/directorcount
2026-09-13 17:26:27,467 INFO client-8090Proxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-13 17:26:29,099 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-13 17:26:29,142 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Sudipta/.staging/job_1789296062547_0002
2026-09-13 17:26:29,586 INFO input.FileInputFormat: Total input files to process : 1
2026-09-13 17:26:29,753 INFO mapreduce.JobSubmitter: number of splits:2
2026-09-13 17:26:30,044 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789296062547_0002
2026-09-13 17:26:30,050 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-13 17:26:30,427 INFO conf.Configuration: resource-types.xml not found
2026-09-13 17:26:30,427 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-13 17:26:30,569 INFO impl.YarnClientImpl: Submitted application application_1789296062547_0002
2026-09-13 17:26:30,680 INFO mapreduce.Job: The url to track the job: http://DESKTOP-AD7D20Q:8088/proxy/application_1789296062547_0002/
2026-09-13 17:26:30,694 INFO mapreduce.Job: Running job: job_1789296062547_0002
2026-09-13 17:26:47,640 INFO mapreduce.Job: Job job_1789296062547_0002 running in uber mode : false
2026-09-13 17:27:04,200 INFO mapreduce.Job: map 0% reduce 0%
2026-09-13 17:27:11,353 INFO mapreduce.Job: map 100% reduce 0%
2026-09-13 17:27:19,491 INFO mapreduce.Job: map 100% reduce 100%
2026-09-13 17:27:29,526 INFO mapreduce.Job: Job job_1789296062547_0002 completed successfully
2026-09-13 17:27:29,732 INFO mapreduce.Job: Counters: 55
  File System Counters
    FILE: Number of bytes read=35639837
    FILE: Number of bytes written=71997021
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
    HDFS: Number of bytes read=172940614
    HDFS: Number of bytes written=9679491
    HDFS: Number of read operations=11
    HDFS: Number of large read operations=0
    HDFS: Number of write operations=2
    HDFS: Number of bytes read erasure-coded=0
  Job Counters
    Killed map tasks=1
    Launched map tasks=3
    Launched reduce tasks=1
    Data-local map tasks=3
    Total time spent by all maps in occupied slots (ms)=38515
    Total time spent by all reduces in occupied slots (ms)=12154
    Total time spent by all map tasks (ms)=38515
    Total time spent by all reduce tasks (ms)=12154
    Total vcore-milliseconds taken by all map tasks=38515
    Total vcore-milliseconds taken by all reduce tasks=12154
    Total megabyte-milliseconds taken by all map tasks=39439360
    Total megabyte-milliseconds taken by all reduce tasks=12445696
  Map-Reduce Framework
    Map input records=2000001
    Map output records=1708822
    Map output bytes=32222187
    Map output materialized bytes=35639843
    Input split bytes=240
    Combine input records=0
    Combine output records=0
    Reduce input groups=569011
    Reduce shuffle bytes=35639843
    Reduce input records=1708822
    Reduce output records=569011
    Spilled Records=3417644
    Shuffled Maps =2
    Failed Shuffles=0
    Merged Map outputs=2
    GC time elapsed (ms)=552
    CPU time spent (ms)=21057
    Physical memory (bytes) snapshot=1168207872
    Virtual memory (bytes) snapshot=1414041600
    Total committed heap usage (bytes)=905969664
    Peak Map Physical memory (bytes)=501268480
    Peak Map Virtual memory (bytes)=568037376
    Peak Reduce Physical memory (bytes)=236494848
    Peak Reduce Virtual memory (bytes)=280936448
  Shuffle Errors
    BAD_ID=0
    CONNECTION=0
    IO_ERROR=0
    WRONG_LENGTH=0
    WRONG_MAP=0
    WRONG_REDUCE=0
  File Input Format Counters
    Bytes Read=172940614
  File Output Format Counters
    Bytes Written=9679491

```

execution1

```

Select Administrator: Command Prompt
Job Counters
  Killed map tasks=1
  Launched map tasks=3
  Launched reduce tasks=1
  Data-local map tasks=3
  Total time spent by all maps in occupied slots (ms)=38515
  Total time spent by all reduces in occupied slots (ms)=12154
  Total time spent by all map tasks (ms)=38515
  Total time spent by all reduce tasks (ms)=12154
  Total vcore-milliseconds taken by all map tasks=38515
  Total vcore-milliseconds taken by all reduce tasks=12154
  Total megabyte-milliseconds taken by all map tasks=39439360
  Total megabyte-milliseconds taken by all reduce tasks=12445696
Map-Reduce Framework
  Map input records=2000001
  Map output records=1708822
  Map output bytes=32222187
  Map output materialized bytes=35639843
  Input split bytes=240
  Combine input records=0
  Combine output records=0
  Reduce input groups=569011
  Reduce shuffle bytes=35639843
  Reduce input records=1708822
  Reduce output records=569011
  Spilled Records=3417644
  Shuffled Maps =2
  Failed Shuffles=0
  Merged Map outputs=2
  GC time elapsed (ms)=552
  CPU time spent (ms)=21057
  Physical memory (bytes) snapshot=1168207872
  Virtual memory (bytes) snapshot=1414041600
  Total committed heap usage (bytes)=905969664
  Peak Map Physical memory (bytes)=501268480
  Peak Map Virtual memory (bytes)=568037376
  Peak Reduce Physical memory (bytes)=236494848
  Peak Reduce Virtual memory (bytes)=280936448
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=172940614
File Output Format Counters
  Bytes Written=9679491

```

Execution2

OUTPUT

cmd Select Administrator: Command Prompt

```

Yoh Komaya      4
Yohahn Ko       2
Yohai Hakak     2
Yohan Begyn     1
Yohan Faure     3
Yohan Guignard  3
Yohan Kurian    4
Yohan Laffort   1
Yohan Leduc     3
Yohan Mabit     1
Yohan Manca     2
Yohan Molles    1
Yohan Riba Martinez  4
Yohan Rozenblat 1
Yohan Rozev     1
Yohan Samel     2
Yohan Shim     1
Yohan Tison     1
Yohan Ungar     6
Yohana Desta   1
Yohana Papa Onyango 1
Yohanan Doron   1
Yohanan Weller  7
Yohance Brinson 2
Yohane Dean Lengol 2
Yohann Abdelnour 1
Yohann Angelvy  1
Yohann Auroux-Bernard 1
Yohann Benedetti 2
Yohann Boennec  2
Yohann Charrin  1
Yohann Compin   1
Yohann Costedoat-Descouzeres 1
Yohann Gloaguen 7
Yohann Grignou  1
Yohann Guglielmetti 1
Yohann Iwandza  3
Yohann Kouam    4
Yohann Sfez     3
Yohann Thiou    2
Yohann Thiriet  1
Yohanna Idha    1
Yohanne Lamoulere 1
Yohannes Feleke 1
Yohannes Yamassee 1
Yohay Hayat     1
Yohei Ishibashi 2
Yohei Ishihama  1
Yohei Kameyama  1

```

output1

```

Zlatko Geleski  3
Zlatko Gotovac  1
Zlatko Grgic    15
Zlatko Kopljari 1
Zlatko Lavanic  4
Zlatko Milojicic 1
Zlatko Pavlinic  7
Zlatko Pavlinovic 1
Zlatko Sacer    1
Zlatko Seselj   1
Zlatko Stojilovic 3
Zlatko Sudovic  13
Zlatko Sviben   3
Zlatko Tomic    9
Zlatko Topcic   1
Zlatolin Donchev 1
Zlatoslava Kryshatafovych 1
Zmacforthewin  1
Zmago Lenardic  1
Zman            1
Zmarai Kasi     1
Zmaster        1
Zmjkk          1
Znere Grace    1
Zo Bell        1
Zo Fan         2
Zo Tahiana Hariminoson 1
Zo Zosak       15
Zoa Zalazar    1
Zoana Price    2
Zobayer Raihan 2
Zobi           1
Zobunus        1
ZodWriter      1
Zodiac Fishgrease 1
Zodwa Nyoni    1
Zoe Aiano      2
Zoe Alker      2
Zoe Allen      1
Zoe Andrikidis 3
Zoe Banks      1
Zoe Barcza     1
Zoe Barnes     1
Zoe Barthelemy 1
Zoe Beck       2
Zoe Bell       1
Zoe Belle Gronner 1
Zoe Beloff     7
Zoe Berriat    7

```

output2

2.3 Job 3: Top 10 Highest Rated Movies

Driver Class: (TopMoviesDriver.java)

```
package topdirectors;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.FloatWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class TopDirectorsDriver {
    public static void main(String[] args) throws Exception {
        if (args.length < 3) {
            System.err.println("Usage: TopDirectorsDriver <input> <tempDir> <output>");
            System.exit(-1);
        }
        // Job 1: Aggregate Average Rating per Director
        Configuration conf1 = new Configuration();
        Job job1 = Job.getInstance(conf1, "DirectorAverageRating");
        job1.setJarByClass(TopDirectorsDriver.class);
        job1.setMapperClass(DirectorAvgMapper.class);
        job1.setReducerClass(DirectorAvgReducer.class);
        job1.setOutputKeyClass(Text.class);
        job1.setOutputValueClass(FloatWritable.class);
        FileInputFormat.addInputPath(job1, new Path(args[0]));
        FileOutputFormat.setOutputPath(job1, new Path(args[1]));
        boolean job1Done = job1.waitForCompletion(true);
        if (job1Done) {
            // Job 2: Sort Descending & Limit to Top 10
            Configuration conf2 = new Configuration();
            Job job2 = Job.getInstance(conf2, "TopDirectorsSort");
            job2.setJarByClass(TopDirectorsDriver.class);
            job2.setMapperClass(DirectorSwapMapper.class);
            job2.setMapOutputKeyClass(FloatWritable.class);
            job2.setMapOutputValueClass(Text.class);
            job2.setSortComparatorClass(DescendingRatingComparator.class);
            job2.setReducerClass(TopNDirectorsReducer.class);
            job2.setOutputKeyClass(Text.class);
            job2.setOutputValueClass(FloatWritable.class);
            job2.setNumReduceTasks(1);
            FileInputFormat.addInputPath(job2, new Path(args[1]));
            FileOutputFormat.setOutputPath(job2, new Path(args[2]));
            System.exit(job2.waitForCompletion(true) ? 0 : 1);
        }
    }
}
```

Mapper Class: (TopMoviesMapper.java)

```
package topmovies;
import java.io.IOException;
import org.apache.hadoop.io.DoubleWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;
```

```
public class TopMoviesMapper
    extends Mapper<Object, Text, DoubleWritable, Text> {
    private final DoubleWritable ratingKey = new DoubleWritable();
    private final Text titleValue = new Text();
    @Override
    public void map(Object key, Text value, Context context)
        throws IOException, InterruptedException {
        String line = value.toString();
        // Skip header
        if (line.startsWith("Title_ID")) {
            return;
        }
        String[] fields = line.split("\t", -1);
        // Need at least 8 columns
        if (fields.length < 8) {
            return;
        }
        String titleId = fields[0].trim();
        String title = fields[1].trim();
        String rating = fields[6].trim();
        String votes = fields[7].trim();
        if (rating.equals("\N") || rating.equals("Unknown")
            || title.equals("\N") || title.isEmpty()) {
            return;
        }
        try {
            double ratingValue = Double.parseDouble(rating);
            String output =
                titleId + "\t" + title + "\tRating=" + ratingValue
                + "\tVotes=" + votes;
            ratingKey.set(ratingValue);
            titleValue.set(output);
            context.write(ratingKey, titleValue);
        } catch (NumberFormatException e) {
            // Ignore invalid records
        }
    }
}
```

Reducer Class: (TopMoviesReducer.java)

```
package topmovies;
import java.io.IOException;
import java.util.PriorityQueue;
import org.apache.hadoop.io.DoubleWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class TopMoviesReducer
    extends Reducer<DoubleWritable, Text, DoubleWritable, Text> {
    private PriorityQueue<MovieRecord> topMovies;
    private static final int TOP_N = 10;
    @Override
    protected void setup(Context context) {
        topMovies = new PriorityQueue<MovieRecord>(
            TOP_N,
            (a, b) -> Double.compare(a.rating, b.rating)
        );
    }
}
```

```
@Override
public void reduce(DoubleWritable key,
    Iterable<Text> values,
    Context context)
    throws IOException, InterruptedException {
    for (Text value : values) {
        MovieRecord movie =
            new MovieRecord(key.get(), value.toString());
        topMovies.offer(movie);
        if (topMovies.size() > TOP_N) {
            topMovies.poll();
        }
    }
}

@Override
protected void cleanup(Context context)
    throws IOException, InterruptedException {
    MovieRecord[] results =
        topMovies.toArray(new MovieRecord[0]);
    java.util.Arrays.sort(
        results,
        (a, b) -> Double.compare(b.rating, a.rating)
    );
    for (MovieRecord movie : results) {
        context.write(
            new DoubleWritable(movie.rating),
            new Text(movie.info)
        );
    }
}

private static class MovieRecord {
    double rating;
    String info;
    MovieRecord(double rating, String info) {
        this.rating = rating;
        this.info = info;
    }
}
}
```

EXECUTION

```
C:\Ashfak\IMDBMoviesAnalysis\topmovies>hadoop jar topmovies.jar topmovies.TopMoviesDriver /imdb_movies/input/imdb_movies.txt /imdb_movies/output/topmovies
2026-09-13 17:43:03,939 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-13 17:43:05,299 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-13 17:43:05,339 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Sudipta/.staging/job_1789296062547_0004
2026-09-13 17:43:05,778 INFO input.FileInputFormat: Total input files to process : 1
2026-09-13 17:43:05,964 INFO mapreduce.JobSubmitter: number of splits:2
2026-09-13 17:43:06,296 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789296062547_0004
2026-09-13 17:43:06,300 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-13 17:43:06,702 INFO conf.Configuration: resource-types.xml not found
2026-09-13 17:43:06,702 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'
2026-09-13 17:43:06,820 INFO impl.VanClientImpl: Submitted application application_1789296062547_0004
2026-09-13 17:43:06,895 INFO mapreduce.Job: The url to track the job: http://DESKTOP-AD7D20Q:8888/proxy/application_1789296062547_0004/
2026-09-13 17:43:06,903 INFO mapreduce.Job: Running job: job_1789296062547_0004
2026-09-13 17:43:22,487 INFO mapreduce.Job: Job job_1789296062547_0004 running in uber mode : false
2026-09-13 17:43:22,492 INFO mapreduce.Job: map 0% reduce 0%
2026-09-13 17:43:41,285 INFO mapreduce.Job: map 50% reduce 0%
2026-09-13 17:43:45,435 INFO mapreduce.Job: map 83% reduce 0%
2026-09-13 17:43:51,673 INFO mapreduce.Job: map 100% reduce 0%
2026-09-13 17:43:59,810 INFO mapreduce.Job: map 100% reduce 100%
2026-09-13 17:44:00,837 INFO mapreduce.Job: Job job_1789296062547_0004 completed successfully
2026-09-13 17:44:01,073 INFO mapreduce.Job: Counters: 55

File System Counters
  FILE: Number of bytes read=212929979
  FILE: Number of bytes written=333658343
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=172940854
  HDFS: Number of bytes written=542
  HDFS: Number of read operations=11
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
  HDFS: Number of bytes read erasure-coded=0

Job Counters
  Killed map tasks=1
  Launched map tasks=3
  Launched reduce tasks=1
  Data-local map tasks=1
  Total time spent by all maps in occupied slots (ms)=49478
  Total time spent by all reduces in occupied slots (ms)=16267
  Total time spent by all map tasks (ms)=49478
  Total time spent by all reduce tasks (ms)=16267
  Total vcore-milliseconds taken by all map tasks=49478
  Total vcore-milliseconds taken by all reduce tasks=16267
  Total megabyte-milliseconds taken by all map tasks=50665472
  Total megabyte-milliseconds taken by all reduce tasks=16657408
```

execution1

```
Map-Reduce Framework
  Map input records=2000001
  Map output records=2000000
  Map output bytes=116008485
  Map output materialized bytes=120011029
  Input split bytes=240
  Combine input records=0
  Combine output records=0
  Reduce input groups=92
  Reduce shuffle bytes=120011029
  Reduce input records=2000000
  Reduce output records=10
  Spilled Records=5551277
  Shuffled Maps=2
  Failed Shuffles=0
  Merged Map outputs=2
  GC time elapsed (ms)=616
  CPU time spent (ms)=26852
  Physical memory (bytes) snapshot=1325338624
  Virtual memory (bytes) snapshot=1577426944
  Total committed heap usage (bytes)=1088946176
  Peak Map Physical memory (bytes)=504123392
  Peak Map Virtual memory (bytes)=572047360
  Peak Reduce Physical memory (bytes)=355848192
  Peak Reduce Virtual memory (bytes)=435884032

Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0

File Input Format Counters
  Bytes Read=172940814
  File Output Format Counters
  Bytes Written=542
```

Execution2

OUTPUT

```
C:\Ashfak\IMDBMoviesAnalysis\topmovies>hdfs dfs -cat /imdb_movies/output/topmovies/part-r-00000
10.0 tt34727714 The Clay Horse Rating=10.0 Votes=26
10.0 tt21065190 Native Renewables: Powering a Nation with the Sun Rating=10.0 Votes=9
10.0 tt29807669 English Rose Rating=10.0 Votes=8
10.0 tt28510081 Eutanazija Rating=10.0 Votes=8
10.0 tt2426280 Catador de sonhos Rating=10.0 Votes=8
10.0 tt28669927 |Ebermensch Rating=10.0 Votes=21
10.0 tt23962784 Internal Rating=10.0 Votes=7
10.0 tt26941897 Bihari lives Matter Rating=10.0 Votes=10
10.0 tt2955724 Losing Senses Rating=10.0 Votes=19
10.0 tt3101358 MILFs Like It Dark Rating=10.0 Votes=6

C:\Ashfak\IMDBMoviesAnalysis\topmovies>
```

Output

2.4 Job 4:Decade-Wise Average Rating

Driver Class: (decadeavg.java)

```
package decadeavg;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.DoubleWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class DecadeDriver {
    public static void main(String[] args) throws Exception {
        if (args.length != 2) {
            System.err.println("Usage: DecadeDriver <input> <output>");
            System.exit(-1);
        }
        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "Decade Wise Average Rating");
        job.setJarByClass(DecadeDriver.class);
        // Mapper
        job.setMapperClass(DecadeMapper.class);
        // Reducer
        job.setReducerClass(DecadeReducer.class);
        // Mapper output
        job.setMapOutputKeyClass(Text.class);
        job.setMapOutputValueClass(DoubleWritable.class);
        // Final output
        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(DoubleWritable.class);
        // Input
        FileInputFormat.addInputPath(job, new Path(args[0]));
        // Output
        FileOutputFormat.setOutputPath(job, new Path(args[1]));
        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

Mapper Class: (DecadeMapper.java)

```
package decadeavg;
import java.io.IOException;
import org.apache.hadoop.io.DoubleWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class DecadeMapper
    extends Mapper<LongWritable, Text, Text, DoubleWritable> {
    private final Text decadeKey = new Text();
    private final DoubleWritable ratingValue = new DoubleWritable();
    @Override
    public void map(LongWritable key, Text value, Context context)
        throws IOException, InterruptedException {
        String line = value.toString();
        // Skip header
```

```
    if (line.startsWith("Title_ID")) {
        return;
    }
    String[] fields = line.split("\t", -1);
    // Need at least 7 columns
    if (fields.length < 7) {
        return;
    }
    String year = fields[3].trim();
    String rating = fields[6].trim();
    if (year.equals("\\N") || year.equals("Unknown")
        || rating.equals("\\N") || rating.equals("Unknown")) {
        return;
    }
    try {
        int yearValue = Integer.parseInt(year);
        double ratingValueDouble = Double.parseDouble(rating);
        int decade = (yearValue / 10) * 10;
        decadeKey.set(decade + "s");
        ratingValue.set(ratingValueDouble);
        context.write(decadeKey, ratingValue);
    } catch (NumberFormatException e) {
        // Ignore invalid records
    }
}
}
```

Reducer Class: (DecadeReducer.java)

```
package decadeavg;
import java.io.IOException;
import org.apache.hadoop.io.DoubleWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class DecadeReducer
    extends Reducer<Text, DoubleWritable, Text, DoubleWritable> {
    private final DoubleWritable result = new DoubleWritable();
    @Override
    public void reduce(Text key, Iterable<DoubleWritable> values,
        Context context)
        throws IOException, InterruptedException {
        double sum = 0.0;
        int count = 0;
        for (DoubleWritable value : values) {
            sum += value.get();
            count++;
        }
        if (count > 0) {
            double average = sum / count;
            // Round to 2 decimal places
            average = Math.round(average * 100.0) / 100.0;
            result.set(average);
            context.write(key, result);
        }
    }
}
```

EXECUTION

```
Select Administrator: Command Prompt

C:\Ashfak\IMDBMoviesAnalysis>decadeavg>hadoop jar decadeavg.jar decadeavg.DecadeDriver /imdb_movies/input/imdb_movies.txt /imdb_movies/output/decadeavg

2026-09-13 17:36:49,128 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-13 17:36:50,631 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-13 17:36:50,675 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Sudipta/.staging/job_1789296062547_0003
2026-09-13 17:36:51,168 INFO input.FileInputFormat: Total input files to process : 1
2026-09-13 17:36:51,341 INFO mapreduce.JobSubmitter: number of splits:2
2026-09-13 17:36:51,700 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789296062547_0003
2026-09-13 17:36:51,702 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-13 17:36:52,132 INFO conf.Configuration: resource-types.xml not found
2026-09-13 17:36:52,148 INFO resource.ResourceUtilits: Unable to find 'resource-types.xml'.
2026-09-13 17:36:52,274 INFO impl.YarnClientImpl: Submitted application application_1789296062547_0003
2026-09-13 17:36:52,370 INFO mapreduce.Job: The url to track the job: http://DESKTOP-A07020Q:8888/proxy/application_1789296062547_0003/
2026-09-13 17:36:52,374 INFO mapreduce.Job: Running job: job_1789296062547_0003
2026-09-13 17:37:07,819 INFO mapreduce.Job: Job job_1789296062547_0003 running in uber mode : false
2026-09-13 17:37:22,335 INFO mapreduce.Job: map 0% reduce 0%
2026-09-13 17:37:25,388 INFO mapreduce.Job: map 100% reduce 0%
2026-09-13 17:37:35,561 INFO mapreduce.Job: map 100% reduce 100%
2026-09-13 17:37:35,577 INFO mapreduce.Job: Job job_1789296062547_0003 completed successfully
2026-09-13 17:37:35,779 INFO mapreduce.Job: Counters: 55
  File System Counters
    FILE: Number of bytes read=31628438
    FILE: Number of bytes written=63974196
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
    HDFS: Number of bytes read=172940854
    HDFS: Number of bytes written=191
    HDFS: Number of read operations=11
    HDFS: Number of large read operations=0
    HDFS: Number of write operations=2
    HDFS: Number of bytes read erasure-coded=0
  Job Counters
    Killed map tasks=1
    Launched map tasks=2
    Launched reduce tasks=1
    Data-local map tasks=2
    Total time spent by all maps in occupied slots (ms)=27527
    Total time spent by all reduces in occupied slots (ms)=9924
    Total time spent by all map tasks (ms)=27527
    Total time spent by all reduce tasks (ms)=9924
    Total vcore-milliseconds taken by all map tasks=27527
    Total vcore-milliseconds taken by all reduce tasks=9924
    Total megabyte-milliseconds taken by all map tasks=28187648
    Total megabyte-milliseconds taken by all reduce tasks=10162176

Map-Reduce Framework
  Map input records=2000000
  Map output records=2000000
  Map output bytes=27628432
  Map output materialized bytes=31628444
  Input split bytes=240
  Combine input records=0
  Combine output records=0
  Reduce input groups=18
  Reduce shuffle bytes=31628444
  Reduce input records=2000000
  Reduce output records=18
  Spilled Records=4000000
  Shuffled Maps =2
  Failed Shuffles=0
  Merged Map outputs=2
  GC time elapsed (ms)=441
  CPU time spent (ms)=15902
  Physical memory (bytes) snapshot=1111613440
  Virtual memory (bytes) snapshot=1368780800
  Total committed heap usage (bytes)=868220928
  Peak Map Physical memory (bytes)=500318208
  Peak Map Virtual memory (bytes)=509438208
  Peak Reduce Physical memory (bytes)=212361216
  Peak Reduce Virtual memory (bytes)=270987264
  Shuffle Errors
    BAD_ID=0
    CONNECTION=0
    IO_ERROR=0
    WRONG_LENGTH=0
    WRONG_MAP=0
    WRONG_REDUCE=0
  File Input Format Counters
    Bytes Read=172940614
  File Output Format Counters
    Bytes Written=191

C:\Ashfak\IMDBMoviesAnalysis>decadeavg>
```

execution1

```
Map-Reduce Framework
  Map input records=2000000
  Map output records=2000000
  Map output bytes=27628432
  Map output materialized bytes=31628444
  Input split bytes=240
  Combine input records=0
  Combine output records=0
  Reduce input groups=18
  Reduce shuffle bytes=31628444
  Reduce input records=2000000
  Reduce output records=18
  Spilled Records=4000000
  Shuffled Maps =2
  Failed Shuffles=0
  Merged Map outputs=2
  GC time elapsed (ms)=441
  CPU time spent (ms)=15902
  Physical memory (bytes) snapshot=1111613440
  Virtual memory (bytes) snapshot=1368780800
  Total committed heap usage (bytes)=868220928
  Peak Map Physical memory (bytes)=500318208
  Peak Map Virtual memory (bytes)=509438208
  Peak Reduce Physical memory (bytes)=212361216
  Peak Reduce Virtual memory (bytes)=270987264
  Shuffle Errors
    BAD_ID=0
    CONNECTION=0
    IO_ERROR=0
    WRONG_LENGTH=0
    WRONG_MAP=0
    WRONG_REDUCE=0
  File Input Format Counters
    Bytes Read=172940614
  File Output Format Counters
    Bytes Written=191

C:\Ashfak\IMDBMoviesAnalysis>decadeavg>
```

Execution2

OUTPUT

```
C:\Ashfak\IMDBMoviesAnalysis\decadeavg>hdfs dfs -cat /imdb_movies/output/decadeavg/part-r-00000
0s      0.0
1870s   5.35
1880s   5.08
1890s   3.08
1900s   0.9
1910s   0.48
1920s   1.2
1930s   2.72
1940s   3.34
1950s   2.87
1960s   2.75
1970s   2.81
1980s   2.91
1990s   2.75
2000s   2.55
2010s   1.66
2020s   1.45
2030s   0.0
```

Output

2.5 Job 5: Year Wise Movies Count

Driver Class: (YearDriver.java)

```
package yearcount;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class YearDriver {
    public static void main(String[] args) throws Exception {
        if (args.length != 2) {
            System.err.println("Usage: YearDriver <input> <output>");
            System.exit(-1);
        }
        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "Year Wise Title Count");
        job.setJarByClass(YearDriver.class);
        job.setMapperClass(YearMapper.class);
        job.setReducerClass(YearReducer.class);
        job.setMapOutputKeyClass(Text.class);
        job.setMapOutputValueClass(IntWritable.class);
        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);
        FileInputFormat.addInputPath(job, new Path(args[0]));
        FileOutputFormat.setOutputPath(job, new Path(args[1]));
        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

Mapper Class: (YearMapper.java)

```
package yearcount;
```

```
import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class YearMapper
    extends Mapper<LongWritable, Text, Text, IntWritable> {
    private final Text yearKey = new Text();
    private final IntWritable one = new IntWritable(1);
    @Override
    public void map(LongWritable key, Text value, Context context)
        throws IOException, InterruptedException {
        String line = value.toString();
        // Skip header
        if (line.startsWith("Title_ID")) {
            return;
        }
        String[] fields = line.split("\t", -1);
        // Need at least 4 columns
        if (fields.length < 4) {
            return;
        }
        // Column 3 = Year
        String year = fields[3].trim();
        if (year.equals("\\N") || year.equals("Unknown")
            || year.isEmpty()) {
            return;
        }
        try {
            Integer.parseInt(year);
            yearKey.set(year);
            context.write(yearKey, one);
        } catch (NumberFormatException e) {
            // Ignore invalid years
        }
    }
}
```

Reducer Class: (YearReducer.java)

```
package yearcount;
import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class YearReducer
    extends Reducer<Text, IntWritable, Text, IntWritable> {
    private final IntWritable result = new IntWritable();
    @Override
    public void reduce(Text key, Iterable<IntWritable> values,
        Context context)
        throws IOException, InterruptedException {
        int total = 0;
        for (IntWritable value : values) {
            total += value.get();
        }
        result.set(total);
    }
}
```

```
context.write(key, result);  
}  
}
```

EXECUTION

```

Select Administrator: Command Prompt
C:\Ashfak\IMDBMoviesAnalysis\yearcount>hadoop jar yearcount.jar yearcount.YearDriver /imdb_movies/input/imdb_movies.txt /imdb_movies/output/yearcount
2026-09-13 17:49:59,673 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-13 17:50:01,019 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-13 17:50:01,050 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Sudipta/.staging/job_1789296062547_0005
2026-09-13 17:50:01,540 INFO input.FileInputFormat: Total input files to process : 1
2026-09-13 17:50:01,727 INFO mapreduce.JobSubmitter: number of splits:2
2026-09-13 17:50:02,057 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789296062547_0005
2026-09-13 17:50:02,061 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-13 17:50:02,428 INFO conf.Configuration: resource-types.xml not found
2026-09-13 17:50:02,429 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-13 17:50:02,588 INFO impl.YarnClientImpl: Submitted application application_1789296062547_0005
2026-09-13 17:50:02,682 INFO mapreduce.Job: The url to track the job: http://DESKTOP-AD7D20Q:8088/proxy/application_1789296062547_0005/
2026-09-13 17:50:02,685 INFO mapreduce.Job: Running job: job_1789296062547_0005
2026-09-13 17:50:17,356 INFO mapreduce.Job: Job job_1789296062547_0005 running in uber mode : false
2026-09-13 17:50:17,356 INFO mapreduce.Job: map 0% reduce 0%
2026-09-13 17:50:33,980 INFO mapreduce.Job: map 50% reduce 0%
2026-09-13 17:50:37,033 INFO mapreduce.Job: map 100% reduce 0%
2026-09-13 17:50:47,322 INFO mapreduce.Job: map 100% reduce 100%
2026-09-13 17:50:49,457 INFO mapreduce.Job: Job job_1789296062547_0005 completed successfully
2026-09-13 17:50:49,911 INFO mapreduce.Job: Counters: 55
File System Counters
  FILE: Number of bytes read=21628438
  FILE: Number of bytes written=43974151
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=172940854
  HDFS: Number of bytes written=1500
  HDFS: Number of read operations=11
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
  HDFS: Number of bytes read erasure-coded=0
Job Counters
  Killed map tasks=1
  Launched map tasks=2
  Launched reduce tasks=1
  Data-local map tasks=2
  Total time spent by all maps in occupied slots (ms)=31064
  Total time spent by all reduces in occupied slots (ms)=10535
  Total time spent by all map tasks (ms)=31064
  Total time spent by all reduce tasks (ms)=10535
  Total vcore-milliseconds taken by all map tasks=31064
  Total vcore-milliseconds taken by all reduce tasks=10535
  Total megabyte-milliseconds taken by all map tasks=31809536
  Total megabyte-milliseconds taken by all reduce tasks=10787840

```

execution1

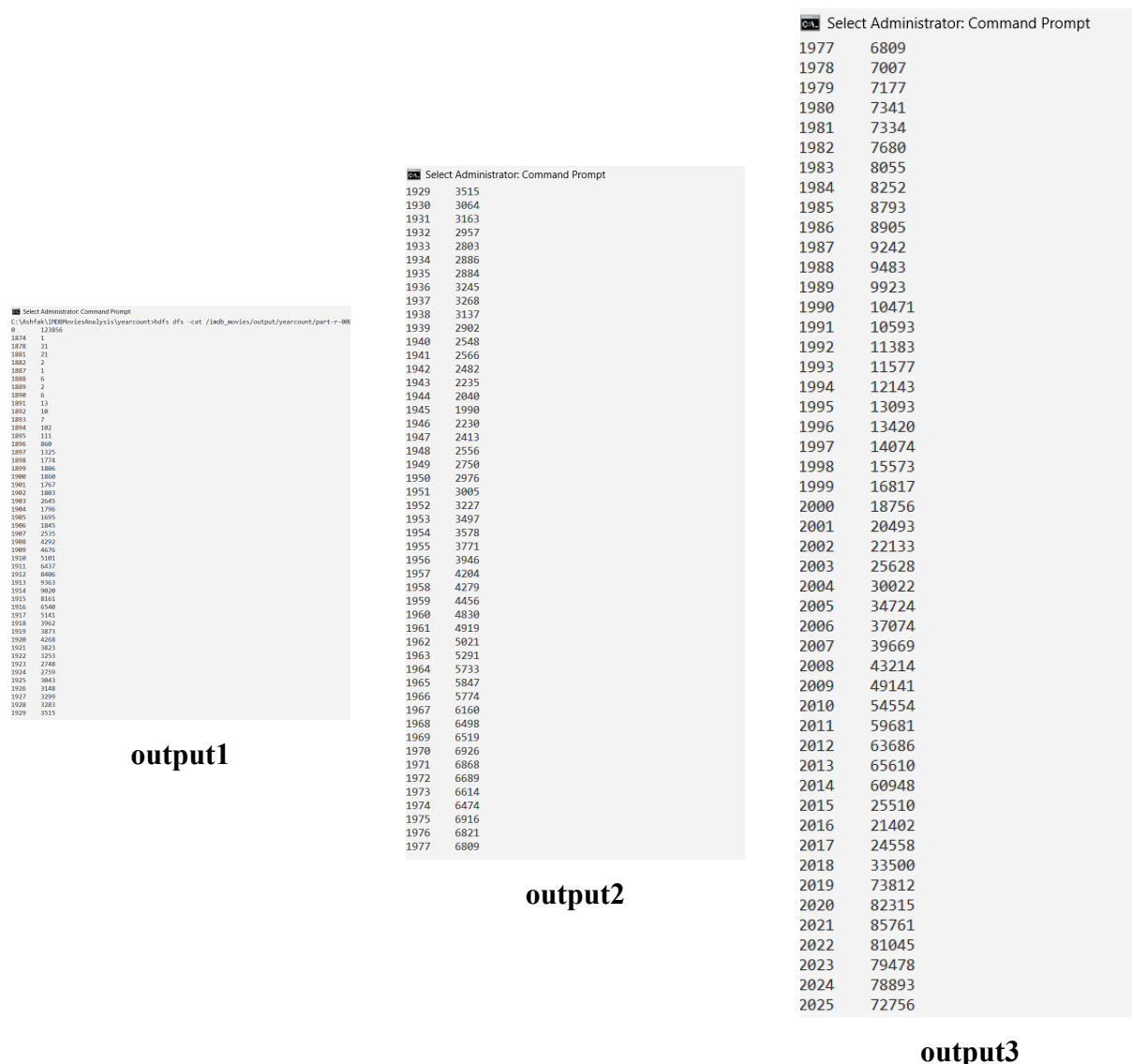
```

Total megabyte-milliseconds taken by all reduce tasks=10787840
Map-Reduce Framework
  Map input records=2000001
  Map output records=2000000
  Map output bytes=17628432
  Map output materialized bytes=21628444
  Input split bytes=240
  Combine input records=0
  Combine output records=0
  Reduce input groups=151
  Reduce shuffle bytes=21628444
  Reduce input records=2000000
  Reduce output records=151
  Spilled Records=4000000
  Shuffled Maps =2
  Failed Shuffles=0
  Merged Map outputs=2
  GC time elapsed (ms)=493
  CPU time spent (ms)=16275
  Physical memory (bytes) snapshot=1178963968
  Virtual memory (bytes) snapshot=1474457600
  Total committed heap usage (bytes)=963641344
  Peak Map Physical memory (bytes)=497491968
  Peak Map Virtual memory (bytes)=569819136
  Peak Reduce Physical memory (bytes)=248569856
  Peak Reduce Virtual memory (bytes)=338735104
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=172940614
File Output Format Counters
  Bytes Written=1500
C:\Ashfak\IMDBMoviesAnalysis\yearcount>

```

Execution2

OUTPUT



3. Apache Pig Analysis

3.1 Job 1: Top 10 Highest-Rated Movies

The movies are sorted according to their ratings in descending order, and the top highest-rated movies are selected.

Script: (top5genres.pig)

```

infile = LOAD 'C:/Ashfak/data/imdb_movies.csv' USING PigStorage(',') AS (
    title_id:chararray, title:chararray, title_type:chararray, year_str:chararray,
    genres:chararray, runtime_str:chararray, rating_str:chararray, votes_str:chararray,
    director_id:chararray, director_name:chararray
);
clean_data = FILTER infile BY title_id != 'Title_ID' AND rating_str MATCHES '^[0-9]+(\\.[0-9]+)?$' AND
    votes_str MATCHES '^[0-9]+$';

```

```

typed_data = FOREACH clean_data GENERATE title_id, title, (float)rating_str AS rating, (int)votes_str AS
    votes, (int)year_str AS year;
credible_movies = FILTER typed_data BY votes >= 50;
ordered_data = ORDER credible_movies BY rating DESC;
top10 = LIMIT ordered_data 10;
STORE top10 INTO 'Top10RatedOutput' USING PigStorage('|');

```

EXECUTION

```

Select Administrator: Command Prompt
HadoopVersion  PigVersion  UserId  StartedAt  FinishedAt  Features
3.2.4  0.18.0  Sudipta  2026-09-13 18:30:11  2026-09-13 18:30:31  ORDER_BY,FILTER,LIMIT

Success!

Job Stats (time in seconds):
JobId  Maps  Reduces  MaxMapTime  MinMapTime  AvgMapTime  MedianMapTime  MaxReduceTime  MinReduceTime  AvgReduceTime  MedianReduceTime  Alias  Feature Outputs
job_local1097204110_0003  1  1  n/a  n/a  n/a  n/a  n/a  n/a  ordered_by_rating  ORDER_BY,COMBINER
job_local152730123_0002  1  1  n/a  n/a  n/a  n/a  n/a  n/a  ordered_by_rating  MAP_ONLY
job_local12142808549_0001  6  0  n/a  n/a  n/a  n/a  0  0  infiles,valid_movies
job_local1682375754_0004  1  1  n/a  n/a  n/a  n/a  n/a  n/a  final_top10,ordered_by_rating  MAP_ONLY
file:///C:/Ashfak/IMDBMoviesAnalysis/pig/Top10RatedOutput,

Input(s):
Successfully read 2000001 records from: "C:/Ashfak/data/imdb_movies.csv"

Output(s):
Successfully stored 10 records in: "file:///C:/Ashfak/IMDBMoviesAnalysis/pig/Top10RatedOutput"

Counters:
Total records written : 10
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local12142808549_0001  ->  job_local152730123_0002,
job_local152730123_0002  ->  job_local1097204110_0003,
job_local1097204110_0003  ->  job_local1682375754_0004,
job_local1682375754_0004

2026-09-13 18:30:31,619 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:30:31,627 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:30:31,627 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:30:31,639 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:30:31,643 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:30:31,643 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:30:31,643 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:30:31,659 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:30:31,661 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:30:31,672 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:30:31,679 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:30:31,683 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:30:31,693 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Encountered Warning FIELD_DISCARDED_TYPE_CONVERSION_FAILED 399319 time(s).
2026-09-13 18:30:31,693 [main] INFO org.apache.pig.Main - Pig script completed in 24 seconds and 452 milliseconds (24452 ms)
2026-09-13 18:30:31,738 [main] INFO org.apache.pig.Main - Pig script completed in 24 seconds and 452 milliseconds (24452 ms)

```

execution1

OUTPUT

```

C:\Ashfak\IMDBMoviesAnalysis>type C:\Ashfak\IMDBMoviesAnalysis\pig\Top10RatedOutput\part-r-00000
t1433175|Romantic Weirdos and the Land of Oddz|2007|Drama|10.0|615
t30429361|Hug of Love 2|2023|Family|10.0|50
t39376578|Joy Awards 2026: Tuba B|Jy|Jk|Jst|Jn|2026|Short|10.0|174
t39305603|Joy Awards 2025: Tuba B|Jy|Jk|Jst|Jn|2026|Short|10.0|246
t0933381|Noel Gallagher: Sitting Here in Silence|2006|Music|9.9|111
t43367796|El Proyecto que Amenaza a M|-xico|2025|Documentary|9.9|80
t0205974|Golpe de Asa|1999|Short|9.9|90
t39144540|Red Sea Film Fest 2024|2025|Short|9.9|1091
t1954538|In Medias Res|2011|Thriller|9.9|770
t0188400|All About Lurleen|1992|Short|9.8|107

C:\Ashfak\IMDBMoviesAnalysis>

```

Output

3.2 Job 2: Movies Count per Genre

The total number of movies is calculated for each genre to identify how many movies belong to each genre.

Script: (genrecount.pig)

```

infiles = LOAD 'C:/Ashfak/data/imdb_movies.csv' USING PigStorage(',') AS (

```

```
title_id:chararray, title:chararray, title_type:chararray, year_str:chararray,
genres:chararray, runtime_str:chararray, rating_str:chararray, votes_str:chararray,
director_id:chararray, director_name:chararray
);
clean_data = FILTER infiles BY title_id != 'Title_ID' AND genres IS NOT NULL AND genres != '';
flattened_genres = FOREACH clean_data GENERATE title_id, FLATTEN(TOKENIZE(genres, ',')) AS genre;
grouped_genre = GROUP flattened_genres BY TRIM(genre);
genre_count = FOREACH grouped_genre GENERATE group AS genre, COUNT(flattened_genres) AS total_movies;
ordered_count = ORDER genre_count BY total_movies DESC;
STORE ordered_count INTO 'GenreCountOutput' USING PigStorage('|');
```

EXECUTION

Select Administrator: Command Prompt

Success!

Job Stats (time in seconds):

JobId	Maps	Reducers	MaxMapTime	MinMapTime	AvgMapTime	MedianMapTime	MaxReduceTime	MinReduceTime	AvgReduceTime	MedianReduceTime	Alias	Feature Outputs
job_local1590748351_0002	1	1	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	ordered_count	SAMPLER
job_local2122686652_0003	1	1	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	ordered_count	ORDER_BY
job_local989876680_0001	6	1	n/a	n/a	n/a	n/a	n/a	n/a	clean_data	flattened_genres,genre_count,grouped_genre,infiles	GROUP_BY,COMBINER	

Input(s):

Successfully read 2000001 records from: "C:/Ashfak/data/imdb_movies.csv"

Output(s):

Successfully stored 1088 records in: "file:///C:/Ashfak/IMDBMoviesAnalysis/pig/GenreCountOutput"

Counters:

Total records written : 1088
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:

job_local989876680_0001 -> job_local1590748351_0002,
job_local1590748351_0002 -> job_local2122686652_0003,
job_local2122686652_0003

2026-09-13 18:50:04,021 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!

2026-09-13 18:50:04,030 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!

2026-09-13 18:50:04,034 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!

2026-09-13 18:50:04,051 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!

2026-09-13 18:50:04,056 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!

2026-09-13 18:50:04,058 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!

2026-09-13 18:50:04,067 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!

2026-09-13 18:50:04,069 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!

2026-09-13 18:50:04,071 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!

2026-09-13 18:50:04,075 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.MapReduceLauncher - Success!

2026-09-13 18:50:04,131 [main] INFO org.apache.pig.Main - Pig script completed in 23 seconds and 482 milliseconds (23482 ms)

execution1

OUTPUT

Select Administrator: Command Prompt

C:\Ashfak\IMDBMoviesAnalysis\pig>type GenreCountOutput\part*

GenreCountOutput\part-r-000000

"Drama|269979
"Comedy|217824
Short|122833
"Documentary|159547
Drama|137733
Documentary|131151
Unknown|95929
Adult|87694
"Action|82013
Comedy|64896
"Animation|62443
"Horror|48377
"Crime|34940
"Short|32436
"Adventure|31007
"Biography|27638
"Music|24947
Music|22221
Horror|21272
Action|21116
"Adult|20550
Thriller|15959
"Fantasy|15295
"Sci-Fi|14611
Sport|14333
"Romance|9530
Romance|9072
"Mystery|9041
"Family|8945
Family|8824
Adventure|8677
Crime|6460
"Musical|5351
Sci-Fi|5269
Animation|4857
Reality-TV|4674
Musical|4642
Fantasy|4312
short|3404
Biography|3455
Mystery|3348
"History|2768
Talk-Show|2747
History|2663

Output

3.3 Job 3: Average Rating per Genre

The average rating of movies is calculated for each year to compare movie ratings across different years.

Script: (genreavg.pig)

```
infile = LOAD 'C:/Ashfak/data/imdb_movies.csv' USING PigStorage(',') AS (
    title_id:chararray, title:chararray, title_type:chararray, year_str:chararray,
    genres:chararray, runtime_str:chararray, rating_str:chararray, votes_str:chararray,
    director_id:chararray, director_name:chararray
);
clean_data = FILTER infile BY title_id != 'Title_ID' AND rating_str MATCHES '^[0-9]+(\\.[0-9]+)?$' AND
    genres IS NOT NULL;
typed_data = FOREACH clean_data GENERATE (float)rating_str AS rating, FLATTEN(TOKENIZE(genres, ',')) AS
    genre;
grouped_genre = GROUP typed_data BY TRIM(genre);
genre_avg = FOREACH grouped_genre GENERATE group AS genre, ROUND_TO(AVG(typed_data.rating), 2) AS
    avg_rating, COUNT(typed_data) AS total_movies;
ordered_avg = ORDER genre_avg BY avg_rating DESC;
STORE ordered_avg INTO 'GenreAvgOutput' USING PigStorage(',');
```

EXECUTION

```
Select Administrator: Command Prompt
success!

ob Stats (time in seconds):
obId  Maps  Reduces MaxMapTime  MinMapTime  AvgMapTime  MedianMapTime  MaxReduceTime  MinReduceTime  AvgReduceTime  MedianReduceTime  Alias  Feature Outputs
ob_local1850665402_0001  6  1  n/a  n/a  n/a  n/a  n/a  n/a  n/a  clean_data,genre_avg,grouped_genre,infiles,typed_data  GROUP_BY,COMBINER
ob_local1255382974_0003  1  1  n/a  n/a  n/a  n/a  n/a  n/a  n/a  ordered_avg  ORDER_BY  file:///C:/Ashfak/IMDBMoviesAnalysis/pig/GenreAvgOutput,
ob_local1265427801_0002  1  1  n/a  n/a  n/a  n/a  n/a  n/a  n/a  ordered_avg  SAMPLER

nput(s):
successfully read 2000001 records from: "C:/Ashfak/data/imdb_movies.csv"

utput(s):
successfully stored 861 records in: "file:///C:/Ashfak/IMDBMoviesAnalysis/pig/GenreAvgOutput"

ounters:
otal records written : 861
otal bytes written : 0
illable Memory Manager spill count : 0
otal bags proactively spilled: 0
otal records proactively spilled: 0

ob DAG:
ob_local1850665402_0001  ->  job_local1265427801_0002,
ob_local1265427801_0002  ->  job_local1255382974_0003,
ob_local1255382974_0003

026-09-13 18:53:18,342 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
026-09-13 18:53:18,342 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
026-09-13 18:53:18,342 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
026-09-13 18:53:18,353 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
026-09-13 18:53:18,353 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
026-09-13 18:53:18,358 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
026-09-13 18:53:18,360 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
026-09-13 18:53:18,360 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
026-09-13 18:53:18,370 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
026-09-13 18:53:18,372 [main] INFO  org.apache.hadoop.mapreduce.executionengine.mapreducelayer.MapReduceLauncher - Success!
026-09-13 18:53:18,419 [main] INFO  org.apache.pig.Main - Pig script completed in 23 seconds and 260 milliseconds (23260 ms)
```

execution1

OUTPUT

```

Select Administrator: Command Prompt

C:\Ashfak\IMDBMoviesAnalysis\pig>type GenreAvgOutput\part*

GenreAvgOutput\part-r-000000

Pigaanod"[2026.0]1
Who Always Goes Braless. - Asakaze Yui"[2026.0]1
Moel"[2026.0]1
Resulting In an NTR (Netorare) Scenario"[2026.0]1
Judith & Tineke Steenbrink - Full Concert"[2026.0]1
Melody Hina Marks"[2026.0]1
and She Sucked and Swallowed My Load Over and Over Again Until I Couldn't Get It to Get Hard Anymore. Nijimura Yumi"[2026.0]1
Slutty Wife Gets Creampied in a Crab-Legged Cowgirl Position."[2026.0]1
Elsewhere"[2026.0]1
BEST 4"[2026.0]1
and Makes Her AV Debut. Track and Field Girl (21)"[2026.0]1
2023"[2026.0]1
25 Years Old"[2026.0]1
Hanging Restraint"[2026.0]1
and She Comes All Over the Place!"[2026.0]1
Mia Kouki"[2026.0]1
and Years"[2026.0]1
Daisies"[2026.0]2
Boca"[2026.0]1
& Murder"[2026.0]1
and Fashionable Artist!"[2026.0]1
Hidden Truth"[2026.0]1
4 Aspiring Idols Who Were Played to Provide Sexual Services After Being Soaked in It"[2026.0]1
Asking for Unprotected Sex!"[2026.0]1
Downtown Date! 37 June Lovejoy"[2026.0]1
divorciada 2"[2025.0]1
and the Day After Tomorrow"[2025.0]1
Revised Edition"[2025.0]1
Lasiodora"[2025.0]1
Aliens and UFOs"[2025.0]1
Making Her Sensitive and Quick to Cum and a Premature Ejaculation Constitution"[2025.0]1
Crete"[2025.0]1
Death"[2025.0]1
an Active Masturbation Pet High School Girl Who's Seen Men Ejaculating Too Much"[2025.0]1
Where You Can Have as Much Sex as You Want! - Melody Hina Marks"[2025.0]1
Lead Development of High School Girls in Their School Life. "[2025.0]1
K-Cup Breasts Massaged in Super-Rich Creampie Three-Shot"[2025.0]1
Cumming a Total of 10 Times in a Row! Kuroshima Rei"[2025.0]1
Facial Cumshot Super Collection Best Uncut 1270Min"[2025.0]1
Flat-Skinned Crossdresser Ino Enoki's AV Debut"[2025.0]1
Very Bad Road Trip"[2025.0]1
les Preyres"[2025.0]1
yiPelea..."[2025.0]1

```

Output

3.4 Job 4: Average Rating & Release Count per Year

Script: (yearavg.pig)

```

infile = LOAD 'C:/Ashfak/data/imdb_movies.csv' USING PigStorage(',') AS (
    title_id:chararray, title:chararray, title_type:chararray, year_str:chararray,
    genres:chararray, runtime_str:chararray, rating_str:chararray, votes_str:chararray,
    director_id:chararray, director_name:chararray
);
clean_data = FILTER infile BY title_id != 'Title_ID' AND year_str MATCHES '^[0-9]{4}$' AND rating_str
    MATCHES '^[0-9]+(\\.[0-9]+)?$';
typed_data = FOREACH clean_data GENERATE (int)year_str AS year, (float)rating_str AS rating;
grouped_year = GROUP typed_data BY year;
year_summary = FOREACH grouped_year GENERATE group AS year, COUNT(typed_data) AS movie_count, ROUND_TO(
    AVG(typed_data.rating), 2) AS avg_rating;
ordered_year = ORDER year_summary BY year DESC;
STORE ordered_year INTO 'YearAvgOutput' USING PigStorage('|');

```

EXECUTION

```
Select Administrator: Command Prompt
Success!

Job Stats (time in seconds):
JobId  Maps  Reduces  MaxMapTime  MinMapTime  AvgMapTime  MedianMapTime  MaxReduceTime  MinReduceTime  AvgReduceTime  MedianReduceTime  Alias  Feature Outputs
job_local1202149297_0002  1  1  n/a  n/a  n/a  n/a  n/a  n/a  n/a  n/a  ordered_year  SAMPLER
job_local146620700_0001  6  1  n/a  n/a  n/a  n/a  n/a  n/a  n/a  n/a  clean_data,grouped_year,infiles,typed_data,year_summary  GROUP_BY,COMBINER
job_local164788502_0003  1  1  n/a  n/a  n/a  n/a  n/a  n/a  n/a  n/a  ordered_year  ORDER_BY  File:///C:/Ashfak/IMDBMoviesAnalysis/pig/YearAvgOutput,

Input(s):
Successfully read 2000001 records from: "C:/Ashfak/data/imdb_movies.csv"

Output(s):
Successfully stored 148 records in: "file:///C:/Ashfak/IMDBMoviesAnalysis/pig/YearAvgOutput"

Counters:
Total records written : 148
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local146620700_0001 -> job_local202149297_0002,
job_local202149297_0002 -> job_local164788502_0003,
job_local164788502_0003

2026-09-13 18:57:22,434 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:57:22,437 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:57:22,437 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:57:22,453 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:57:22,453 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:57:22,453 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:57:22,463 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:57:22,465 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:57:22,469 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:57:22,478 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-13 18:57:22,514 [main] INFO org.apache.pig.Main - Pig script completed in 24 seconds and 440 milliseconds (24440 ms)

C:\Ashfak\IMDBMoviesAnalysis\pig>
```

execution1

OUTPUT

```
Select Administrator: Command Prompt

C:\Ashfak\IMDBMoviesAnalysis\pig>type YearAvgOutput\part*

YearAvgOutput\part-r-000000

2032|1|0.0
2030|6|3.17
2029|11|0.91
2028|36|3.33
2027|567|3.24
2026|25363|10.62
2025|62032|10.15
2024|67942|9.97
2023|60766|10.31
2022|70191|9.56
2021|74736|8.15
2020|72205|7.63
2019|64333|7.78
2018|29663|6.59
2017|21468|6.89
2016|17983|8.18
2015|10900|12.29
2014|39942|12.67
2013|42977|12.34
2012|43078|11.86
2011|40573|12.16
2010|36929|12.63
2009|37270|12.47
2008|36122|11.38
2007|33840|11.26
2006|31810|11.43
2005|30047|11.32
2004|25999|12.07
2003|22040|12.91
2002|18670|13.52
2001|17416|12.78
2000|16031|13.31
1999|14426|13.2
1998|13356|13.4
1997|11956|14.23
1996|11411|14.26
1995|11136|14.31
1994|10207|14.42
1993|9869|15.28
1992|9772|14.81
1991|8977|16.1
1990|8821|15.75
1989|8347|17.22
```

Output

 Select Administrator: Command Prompt

```
1929|2921|9.59
1928|2806|10.84
1927|2788|10.98
1926|2592|11.62
1925|2538|9.52
1924|2341|9.21
1923|2333|8.04
1922|2729|7.7
1921|3308|6.21
1920|3759|5.51
1919|3473|6.06
1918|3496|6.99
1917|4450|5.27
1916|5649|3.22
1915|7106|2.34
1914|7759|2.56
1913|8102|3.61
1912|7053|1.46
1911|5657|1.14
1910|4555|0.87
1909|4254|0.7
1908|3853|0.61
1907|2242|0.61
1906|1603|0.99
1905|1335|0.69
1904|1504|0.58
1903|2219|0.62
1902|1564|0.42
1901|1453|0.46
1900|1623|0.53
1899|1478|0.5
1898|1480|0.71
1897|1102|1.06
1896|686|1.13
1895|84|1.72
1894|81|2.47
1893|7|1.74
1892|9|2.86
1891|9|2.9
1890|2|1.0
1889|1|1.0
1888|5|1.72
1887|1|1.0
1882|2|1.0
1881|21|4.57
1878|30|1.0
```

```
YearAvgOutput\.part-r-00000.crc
```

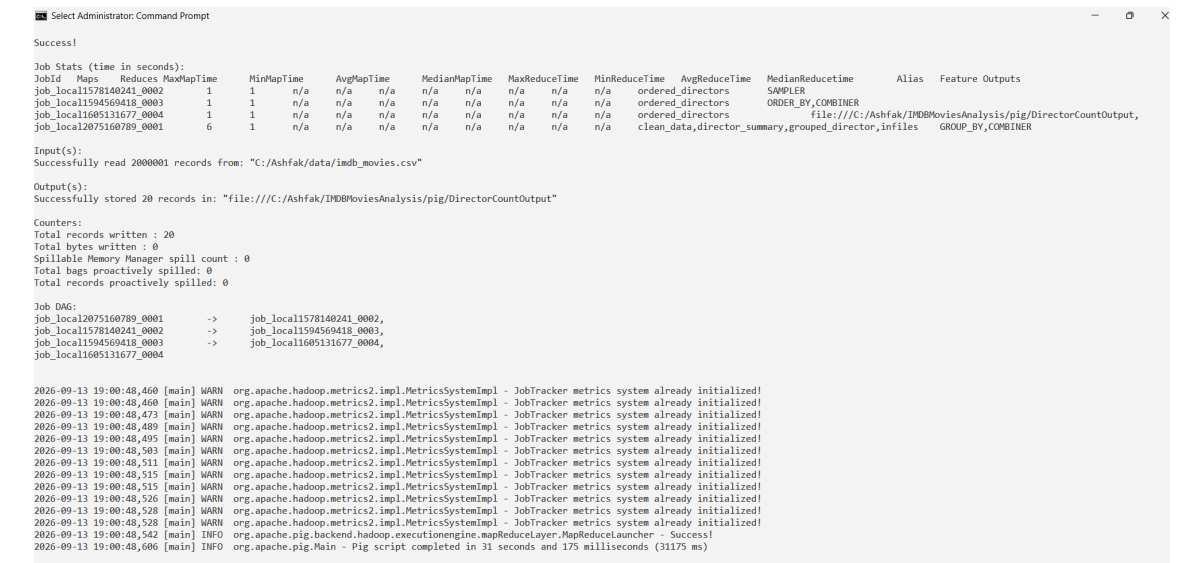
Output

3.5 Job 5: Top Directors by Movie Count

Script: (directorcount.pig)

```
infile = LOAD 'C:/Ashfak/data/imdb_movies.csv' USING PigStorage(',') AS (
    title_id:chararray, title:chararray, title_type:chararray, year_str:chararray,
    genres:chararray, runtime_str:chararray, rating_str:chararray, votes_str:chararray,
    director_id:chararray, director_name:chararray
);
clean_data = FILTER infile BY title_id != 'Title_ID' AND director_name IS NOT NULL AND director_name !=
    '' AND director_name != '\\N';
grouped_director = GROUP clean_data BY director_name;
director_summary = FOREACH grouped_director GENERATE group AS director_name, COUNT(clean_data) AS
    total_movies;
ordered_directors = ORDER director_summary BY total_movies DESC;
top_directors = LIMIT ordered_directors 20;
STORE top_directors INTO 'DirectorCountOutput' USING PigStorage('|');
```

EXECUTION



execution1

OUTPUT

```

C:\Ashfak\IMDBMoviesAnalysis\pig>type DirectorCountOutput\part*

DirectorCountOutput\part-r-00000

Unknown|262470
0|184246
9|3370
8|3264
10|3220
7|3183
11|2968
12|2740
13|2635
14|2376
6|2354
15|2212
16|2085
18|1874
17|1872
19|1680
20|1615
21|1524
22|1451
5|1319

DirectorCountOutput\.part-r-00000.crc
ccac01

```

Output

3.6 Job 6: Content Count by Title Type

Script: (typecount.pig)

```

infile = LOAD 'C:/Ashfak/data/imdb_movies.csv' USING PigStorage(',') AS (
  title_id:chararray, title:chararray, title_type:chararray, year_str:chararray,
  genres:chararray, runtime_str:chararray, rating_str:chararray, votes_str:chararray,
  director_id:chararray, director_name:chararray
);
clean_data = FILTER infile BY title_id != 'Title_ID' AND title_type IS NOT NULL;
grouped_type = GROUP clean_data BY title_type;
type_summary = FOREACH grouped_type GENERATE group AS title_type, COUNT(clean_data) AS total_titles;
ordered_type = ORDER type_summary BY total_titles DESC;
STORE ordered_type INTO 'TypeCountOutput' USING PigStorage('|');

```

EXECUTION

```

Select Administrator: Command Prompt
Success!

Job Stats (time in seconds):
JobId  Maps  Reduces  MaxMapTime  MinMapTime  AvgMapTime  MedianMapTime  MaxReduceTime  MinReduceTime  AvgReduceTime  MedianReduceTime  Alias  Feature Outputs
job_local1035371738_0001  6  0  n/a  n/a  n/a  n/a  0  0  0  clean_data,infiles,popular_hits,typed_data  MAP_ONLY
job_local1235722855_0004  1  1  n/a  n/a  n/a  n/a  n/a  n/a  n/a  ordered_hits  file:///C:/Ashfak/IMDBMoviesAnalysis/pig/HighRatedPopularOutput,
job_local1882535059_0003  1  1  n/a  n/a  n/a  n/a  n/a  n/a  n/a  ordered_hits  ORDER_BY,COMBINER
job_local1684680715_0002  1  1  n/a  n/a  n/a  n/a  n/a  n/a  ordered_hits  SAMPLER

Input(s):
Successfully read 2000001 records from: "C:/Ashfak/data/imdb_movies.csv"

Output(s):
Successfully stored 20 records in: "file:///C:/Ashfak/IMDBMoviesAnalysis/pig/HighRatedPopularOutput"

Counters:
Total records written : 20
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local1035371738_0001  ->  job_local1684680715_0002,
job_local1684680715_0002 ->  job_local1882535059_0003,
job_local1882535059_0003 ->  job_local1235722855_0004,
job_local1235722855_0004

2026-09-13 19:14:43,230 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:14:43,237 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:14:43,239 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:14:43,245 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:14:43,245 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:14:43,245 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:14:43,257 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:14:43,258 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:14:43,268 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:14:43,266 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:14:43,268 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:14:43,273 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:14:43,282 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Encountered Warning FIELD_DISCARDED_TYPE_CONVERSION_FAILED 16613 time(s).
2026-09-13 19:14:43,282 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-13 19:14:43,337 [main] INFO org.apache.pig.Main - Pig script completed in 17 seconds and 709 milliseconds (17709 ms)

```

execution1

OUTPUT

```
C:\Ashfak\IMDBMoviesAnalysis\pig>type HighRatedPopularOutput\part*
```

```
HighRatedPopularOutput\part-r-00000
```

```

The Shawshank Redemption|9.3|3233087|1994
Batman Begins|8.2|1741893|2005
American Beauty|8.3|1287445|1999
One Flew Over the Cuckoo's Nest|8.6|1165820|1975
Raiders of the Lost Ark|8.4|1119851|1981
Requiem for a Dream|8.3|983780|2000
Indiana Jones and the Last Crusade|8.2|870773|1989
Gran Torino|8.1|869501|2008
There Will Be Blood|8.2|718146|2007
Rain Man|8.0|589161|1988
The Kashmir Files|8.5|579811|2022
The Help|8.1|532629|2011
The Exorcist|8.1|504094|1973
Monty Python's Life of Brian|8.0|435717|1979
The Hunt|8.3|417446|2012
Scent of a Woman|8.0|375708|1992
Magnolia|8.0|352787|1999
A Separation|8.3|280196|2011
Bicycle Thieves|8.2|193133|1948
Network|8.0|187504|1976

```

```
HighRatedPopularOutput\.part-r-00000.crc
```

```
crCXmVoo|i.~
```

```
C:\Ashfak\IMDBMoviesAnalysis\pig>
```

Output

3.7 Job 7: Bottom 20 Lowest-Rated Titles

Script: (bottomrated.pig)

```

infile = LOAD 'C:/Ashfak/data/imdb_movies.csv' USING PigStorage(',') AS (
    title_id:chararray, title:chararray, title_type:chararray, year_str:chararray,
    genres:chararray, runtime_str:chararray, rating_str:chararray, votes_str:chararray,
    director_id:chararray, director_name:chararray
);
-- Clean data and cast types safely

```

```

clean_data = FILTER infiles BY
  title_id != 'Title_ID' AND
  rating_str MATCHES '^[0-9]+(\\.[0-9]+)?$' AND
  votes_str MATCHES '^[0-9]+$';
typed_data = FOREACH clean_data GENERATE
  title_id,
  title,
  (int)year_str AS year,
  genres,
  (float)rating_str AS rating,
  (int)votes_str AS votes;
-- Ensure minimum vote threshold for credibility
credible_movies = FILTER typed_data BY votes >= 50;
-- Sort by rating ascending (lowest first)
ordered_bottom = ORDER credible_movies BY rating ASC, votes DESC;
bottom20_rated = LIMIT ordered_bottom 20;
STORE bottom20_rated INTO 'Bottom20RatedOutput' USING PigStorage('|');

```

EXECUTION

```

Select Administrator: Command Prompt
Success!

Job Stats (time in seconds):
JobId  Maps  Reduces  MaxMapTime  MinMapTime  AvgMapTime  MedianMapTime  MaxReduceTime  MinReduceTime  AvgReduceTime  MedianReduceTime  Alias  Feature Outputs
job_local1177611623_0004  1  1  n/a  n/a  n/a  n/a  n/a  n/a  n/a  n/a  ordered_bottom
job_local1468242486_0002  1  1  n/a  n/a  n/a  n/a  n/a  n/a  n/a  n/a  ordered_bottom  SAMPLER
job_local2020351085_0001  6  0  n/a  n/a  n/a  n/a  0  0  0  0  clean_data,credible_movies,infiles,typed_data  MAP_ONLY
job_local1818631992_0003  1  1  n/a  n/a  n/a  n/a  n/a  n/a  n/a  n/a  ordered_bottom  ORDER_BY,COMBINER

Input(s):
Successfully read 2000001 records from: "C:/Ashfak/data/imdb_movies.csv"

Output(s):
Successfully stored 20 records in: "file:///C:/Ashfak/IMDBMoviesAnalysis/pig/Bottom20RatedOutput"

Counters:
Total records written : 20
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local2020351085_0001  ->  job_local1468242486_0002,
job_local1468242486_0002  ->  job_local1818631992_0003,
job_local1818631992_0003  ->  job_local1177611623_0004,
job_local1177611623_0004

2026-09-13 19:07:47,565 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:07:47,564 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:07:47,586 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:07:47,595 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:07:47,597 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:07:47,600 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:07:47,604 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:07:47,606 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:07:47,608 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:07:47,618 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:07:47,621 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:07:47,621 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:07:47,625 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:07:47,627 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-13 19:07:47,687 [main] INFO org.apache.pig.Main - Pig script completed in 24 seconds and 699 milliseconds (24699 ms)

```

execution1

OUTPUT

```

C:\Ashfak\IMDBMoviesAnalysis\pig>type Bottom20RatedOutput\part*

Bottom20RatedOutput\part-r-00000

tt13423846|321 Action|2020|Drama|1.0|10341
tt37836776|Kadik|y Bogasi|2025|Comedy|1.0|7044
tt1852818|Justin Bieber: Rise to Fame|2011|Documentary|1.0|2845
tt15443254|Cumali Ceber 666|2022|Comedy|1.0|1190
tt11529726|Laz Kit|2020|Comedy|1.0|963
tt32125636|Donald Trump: Made in the USA|2024|Unknown|1.0|810
tt22485534|Think Big and Kick Ass in Business and Life|2007|Unknown|1.0|436
tt0334193|QIK2J0G|2002|Short|1.0|294
tt13232730|Hearts Are Trump|2020|Drama|1.0|256
tt10642850|15 Temmuz destani|2019|News|1.0|217
tt29868945|Dressing Like Meghan Markle|2023|Documentary|1.0|203
tt0884106|Jocking Around|2007|Documentary|1.0|191
tt1392266|Who's Watching Who?|2000|Comedy|1.0|182
tt1872215|Tunnel Rendez-vous|2011|Comedy|1.0|155
tt21938460|Trip to the Moon|2007|Short|1.0|148
tt12280046|Queer & Fier(ce)|2019|Documentary|1.0|103
tt0374100|Onnanoko no heya: O-teire namachuukei!!|1998|Unknown|1.0|67
tt1914390|Toukyou doutei|2010|Short|1.0|66
tt0310048|Reality|1969|Documentary|1.0|66
tt0196388|Blood of the Undead: The Unwanted|1987|Short|1.0|59

```

Output